ModelIt!

Teacher's Guide

Solar vs. Fossil: The Energy Showdown

Which Power Source Wins — and What's the Tradeoff?

Quick Reference

Grade Level	Grade 05L2
NGSS Standard	5-PS3-1, 5-ESS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: 5-PS3-1, 5-ESS3-1

Use models to describe that energy in animals' food was once energy from the sun. Obtain and combine information about ways individual communities use science ideas to protect the Earth's resources and environment.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model comparing solar and fossil fuel energy systems and their environmental impacts.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems** — Students model how energy choices affect atmospheric CO2 and air quality.
Crosscutting Concept	**Cause and Effect** — Students identify causal relationships between energy source choices and environmental outcomes.

CAST Testing Alignment

MULTIPLE CHOICE:

A city switches from coal power plants to solar panels. Which of the following will most likely decrease?

CONSTRUCTED RESPONSE:

Explain why a town cannot rely on solar energy alone and describe one solution to this problem.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Humans need energy to power homes, schools, and transportation. The two main sources — fossil fuels and solar energy — have very different impacts on the environment.

How Fossil Fuels Work

Coal, oil, and natural gas are burned to heat water into steam, which spins turbines to generate electricity. This process releases CO2 and other pollutants into the atmosphere.

How Solar Panels Work

Solar panels contain cells that convert sunlight directly into electricity. No burning is involved, so no CO2 is released during operation.

The Air Quality Connection

CO2 and particulate matter from burning fossil fuels reduce air quality. Poor air quality causes respiratory problems, especially for children and elderly populations.

The Reliability Tradeoff

Fossil fuels can generate power 24/7, rain or shine. Solar panels only produce energy during daylight hours and are affected by clouds, seasons, and latitude.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Sunlight Hours, Fossil Fuel Burn Rate, Solar Energy Output, CO2 Emissions, and Air Quality Index as the five system components.

E **Establish Relationships**
Students determine that Sunlight drives Solar Output (+), Fossil Fuels drive CO2 Emissions (+), CO2 reduces Air Quality (-), and Solar Output reduces the need for fossil fuels, thereby reducing emissions (-).

V **Visualize & Model**
Students build the energy comparison model showing how two energy sources produce different environmental outcomes.

E **Evaluate Outcomes**
Students run All Solar and Cloudy Day scenarios to discover the strengths and weaknesses of each energy source.

R **Revise & Extend**
Students add Energy Storage (batteries) or Wind Energy to explore how the system can be improved.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Two ways to make electricity. One is clean. One is not. Which should we choose?"*

Watch For

Show split image and ask students to identify what they see on each side.

Slide 2: Learning Objectives

2 min

Say: *"Today we model the energy showdown between solar and fossil fuels."*

Watch For

Students read objectives and predict which energy source is better.

Slide 3: Big Question

2 min

Say: *"This is not a simple yes or no. Our model will reveal the tradeoffs."*

Watch For

Quick poll: Who thinks solar can replace fossil fuels completely?

Slide 4: LEVER Framework

2 min

Say: *"We will build a model that compares both energy sources side by side."*

Watch For

Preview how the model will have two inputs and three outputs.

Slide 5: Activity 1: Components

8 min

Say: *"Our model has two external inputs: Sunlight Hours and Fossil Fuel Burn Rate. The three internal components respond to these inputs."*

Watch For

Students sort components and discuss which are controllable.

Slide 6: Activity 2: Connections

8 min

Say: *"Notice that solar energy has an INDIRECT negative effect on emissions — by replacing fossil fuels."*

Watch For

Guide students through drawing all relationship arrows.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test what happens when we go all solar — and then when clouds roll in."*

Watch For

Students run All Solar and Cloudy Day scenarios and compare results.

Slide 8: Discoveries

5 min

Say: *"Solar is cleaner, but fossil fuels are more reliable. That's the tradeoff."*

Watch For

Class discussion: What's the best mix?

Slide 9: STEM Challenge

10 min

Say: *"Now design an energy plan that cuts emissions by 50% without blackouts."*

Watch For

Groups create plans; circulate to challenge assumptions.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Sunlight Hours, Fossil Fuel Burn Rate
- **INTERNAL:** Solar Energy Output, CO2 Emissions, Air Quality Index

Relationships

- **Sunlight Hours to Solar Energy Output:** POSITIVE (+) — More hours of sunlight means more electricity generated from solar panels.
- **Fossil Fuel Burn Rate to CO2 Emissions:** POSITIVE (+) — Burning more fossil fuel releases more CO2 into the atmosphere.
- **CO2 Emissions to Air Quality Index:** NEGATIVE (-) — More CO2 and pollutants in the air means worse air quality.
- **Solar Energy Output to CO2 Emissions:** NEGATIVE (-) — When solar provides more energy, less fossil fuel is needed, reducing emissions.

Key Discoveries

- Solar energy produces NO CO2 emissions — it is completely clean
- Fossil fuels produce reliable energy but release CO2 that harms air quality
- Solar energy depends on sunlight — cloudy days and nighttime are a problem
- Replacing fossil fuels with solar improves air quality but requires backup for low-sun periods

Multiple Choice Post-Assessment Answer Key

1. In the energy model, what happens to CO2 Emissions when Fossil Fuel Burn Rate is set to 0% and Sunlight Hours are at 80%?

Correct: C) CO2 Emissions drop to near zero because solar energy produces no CO2

Correct! Solar panels convert sunlight directly into electricity without burning anything. No burning means no CO2 emissions. When fossil fuel use drops to zero, emissions drop to near zero.

2. What is the main tradeoff between solar energy and fossil fuel energy revealed by the model?

Correct: C) Solar energy is clean but depends on sunlight; fossil fuels are reliable but produce CO2 that harms air quality

That's right! The model shows that solar energy produces zero emissions but only works when the sun is shining. Fossil fuels work anytime but release CO2 that reduces air quality. This is the core tradeoff.

3. What type of relationship exists between Fossil Fuel Burn Rate and Air Quality Index?

Correct: C) Negative: burning more fossil fuels releases CO2 and pollutants that reduce air quality

Correct! Burning fossil fuels releases CO2 and particulate matter into the atmosphere. More burning means more pollution, which lowers the Air Quality Index. This is an indirect negative relationship through CO2 Emissions.

4. During the Cloudy Day Crisis scenario (Sunlight at 10%, Fossil Fuel at 0%), what problem does the model reveal?

Correct: B) Solar Energy Output crashes and very little total energy is produced

Correct! With very little sunlight and no fossil fuels, almost no electricity is generated. This reveals the reliability problem of solar-only energy: the community would face blackouts on cloudy days or at night without backup or storage.

5. A city switches from 100% fossil fuels to 50% solar and 50% fossil fuels. Based on the model, what would you predict?

Correct: C) CO_2 Emissions would decrease by about half, and Air Quality would improve

That's right! Cutting fossil fuel use in half means roughly half the CO_2 emissions. Less CO_2 means better air quality. The solar energy fills in the gap without adding any emissions.

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Solar panels work at night"	Solar panels require sunlight to generate electricity. At night, stored energy from batteries or other sources must be used.
"Fossil fuels are not related to air quality"	Burning fossil fuels releases CO2, sulfur dioxide, and particulate matter directly into the air, reducing air quality.
"One energy source is always better"	Both solar and fossil fuels have advantages and disadvantages. The best solution depends on location, climate, technology, and community needs.